Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

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***Abstract*—We study online control of a scalar storage system with nonnegative adversarial resource arrivals and state-dependent action constraints. The cost is adversarial, time-varying, and depends on both the action and the state. At each time, the controller selects a feasible action before the current resource arrival and cost are revealed. To handle the resulting coupling between feasibility and online learning, we introduce *Simplex Disturbance-Action Control* (SDAC), an H-dimensional finite-memory policy class parameterized by a subprobability simplex. The key feature of SDAC is that every fixed policy in the class is feasible by construction. Our adaptive controller updates these parameters by entropic online mirror descent while preserving feasibility along the time-varying trajectory. We prove regret $O\left(\sqrt{T\log(H+1)}\right)$ relative to the best fixed SDAC policy. As the retention coefficient α approaches one, the leading factor scales as $\Theta((1-\alpha)^{-3/2})$. We also derive a minimax lower bound which, when T is at least on the order of $(1-\alpha)^{-1}$, scales as $\Omega((1-\alpha)^{-1}\sqrt{T})$ near $\alpha=1$, leaving a factor $(1-\alpha)^{-1/2}$ between the upper and lower bounds. We also show that SDAC geometrically approximates a decreasing-allocation policy class that spreads each arrival over time according to a common decreasing allocation sequence, a structure that arises in stochastic energy-harvesting control. Geometric allocation sequences induce all feasible proportional state-feedback policies. With $H=\Theta(\log T)$, the resulting regret against the decreasing-allocation class, and hence against feasible proportional state-feedback policies, is $O\left(\sqrt{T\log\log T}\right)$. The framework applies to energy-harvesting batteries and other storage-constrained resource systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state-dependent action constraints.**

I. Introduction

A. Motivation and Problem Setting

MANY resource-allocation systems have a storage state that changes over time and limits the control action at each time. An action that lowers the current cost may leave less resource for future decisions, while conserving the resource may increase the current cost. The controller must balance these effects while satisfying the action constraint at every time.

We study this problem for a scalar storage system that loses a fixed fraction of its stored resource each period. At each time, the controller observes the current state and all past arrivals and cost functions, chooses a control action, and then observes the current arrival and cost function. The arrival and cost sequences may be chosen adversarially based on past observations; we do not assume a known probability model. Our goal is to construct a feasible online controller whose cumulative cost is close to that of the best fixed policy, chosen in hindsight, from a specified benchmark class.

The main difficulty is that the current action affects both the next state and the actions that will be feasible later. Therefore, the controller cannot update its decisions independently at each time. *Disturbance-Action Control* (DAC) policies address this dependence by combining a stabilizing state-feedback term with a linear function of past disturbances. In our storage model, however, unrestricted filter weights can produce a negative action or an action larger than the available resource.

We address this issue by introducing *Simplex Disturbance-Action Control* (SDAC). SDAC restricts the filter weights to a subprobability simplex and includes powers of the retention coefficient in the filter. These choices ensure that every fixed SDAC policy satisfies the action constraint. The induced state and action are also affine in the policy parameter, so a convex state–action cost is convex as a function of that parameter. The parameter can therefore be updated using online convex optimization.

B. Model and Information Pattern

The storage state $x_t \geq 0$ evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t,$$

where $\alpha \in (0,1)$ is a fixed retention coefficient, u_t is the control action, and $w_t \in [0,1]$ is a nonnegative resource arrival. The action must satisfy

$$0 \leq u_t \leq \alpha x_t.$$

Thus, the controller cannot use more than the amount that remains after the storage loss.

At time t, the controller chooses u_t using the current state and the information available through time $t-1$. The arrival w_t and the convex cost function c_t are revealed after the action is selected, and the controller incurs the cost $c_t(x_t, u_t)$. The pair (w_t, c_t) may depend on the observations through time $t-1$, but not on the current action. Both the online trajectory and every benchmark trajectory must satisfy the same dynamics and action constraint. The formal model and assumptions are given in Section III.

Energy-harvesting battery management is our main application. In this case, x_t is the battery state of charge, w_t is the harvested energy, u_t is the energy used, and $1-\alpha$ is the fraction of stored energy lost during one period. Similar dynamics arise in perishable inventory systems, water reservoirs with

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storage loss, and queues with abandonment. In many standard queueing models, $\alpha = 1$; that case is outside our main results and is discussed in Section VII.

C. Contributions

Our contributions are as follows:

- **Policy class.** We introduce SDAC, a finite-memory policy class parameterized by a subprobability simplex. We prove that every fixed SDAC policy is feasible for every arrival sequence with entries in $[0, 1]$. We also show that the state and action generated by a fixed SDAC policy are affine in its parameter. Under our convexity and Lipschitz assumptions on the state–action costs, the induced cost is convex and Lipschitz as a function of the policy parameter.
  We further show that SDAC approximates the decreasing-allocation policy class defined in Section IV-C. This class allocates each arrival over future periods according to the same decreasing sequence. It contains every feasible proportional state-feedback policy $u_t = kx_t$, with $k \in [0, \alpha]$, and the approximation error decreases geometrically with the memory horizon H.
- **Online control and regret.** We develop an online controller that updates its SDAC parameter using entropic online mirror descent (OMD) over the subprobability simplex. Because the parameter changes over time, the nominal SDAC action is clipped when needed to satisfy the current action constraint. We prove that this controller remains feasible and achieves
  $$O\Big(\sqrt{T \log(H + 1)}\Big)$$
  regret against the best fixed SDAC policy; see Theorem 1. For fixed L_x and L_u, the leading factor scales as $\Theta((1 - \alpha)^{-3/2})$ as $\alpha \uparrow 1$.
  For fixed α, take the memory horizon H to grow as $\log T$. The resulting regret is $O(\sqrt{T \log \log T})$ against the decreasing-allocation class. The same bound holds against the best feasible proportional state-feedback policy; see Corollaries 2 and 3.
- **Minimax lower bound.** We prove a lower bound of order $\sqrt{T}$ on the expected regret of every causal feasible controller, including randomized controllers, even when the arrival and cost sequences are deterministic and fixed in advance; see Theorem 2. When T is at least on the order of $(1 - \alpha)^{-1}$, the lower bound scales as $\Omega((1 - \alpha)^{-1}\sqrt{T})$ near $\alpha = 1$; see Corollary 1. The dependence of the upper and lower bounds on $1 - \alpha$ therefore differs by a factor $(1 - \alpha)^{-1/2}$. Closing this gap is left for future work.

D. Related Work

a) Online nonstochastic control: Online nonstochastic control studies dynamical systems with adversarial disturbances and costs. Performance is measured by regret against a policy chosen in hindsight, under suitable stability conditions [1], [2]; see [3] for an overview. For known linear systems, Simchowitz et al. establish the optimal $\sqrt{T}$ dependence on the horizon while leaving some dependence on the system's stability unresolved [4]. Related work also considers partial observability [5] and bandit feedback [6].

A central idea in this literature is DAC. A DAC policy combines a stabilizing state-feedback action with a finite linear filter of past disturbances. The action is therefore described by finitely many policy parameters that can be updated using online methods. SDAC uses the same basic idea but restricts the parameters in a way suited to storage systems with nonnegative arrivals.

Online control with affine or more general constraints on states and actions has been studied in [7], [8]. The constraint $u_t \leq \alpha x_t$ is affine in the state and action, but the range of allowed actions depends on the current storage state. Our approach uses the structure of the storage dynamics to ensure that every fixed SDAC policy is feasible and that the online controller remains feasible while its parameter changes.

Our analysis also considers how regret depends on the retention coefficient α. Kumar et al. study online convex optimization in which past decisions can affect all later costs, and derive bounds that depend on how quickly those effects decay [9]. In their model with geometrically decaying memory, the lower bound scales as $\sqrt{T}$ divided by one minus the decay factor, paralleling our $\Omega(\sqrt{T}/(1 - \alpha))$ lower bound for sufficiently long horizons. Our lower bound is proved directly for state–action costs under the state-dependent action constraint.

b) Energy harvesting: Several energy-harvesting policies adjust energy use according to the available battery level. Shaviv and Özgür [10] study a policy that uses a fixed fraction of the available energy and prove guarantees under independent and identically distributed (i.i.d.) energy arrivals. Arafa et al. [11] consider more general sequences for spending energy after a recharge. Zibaeenejad and Chen [12] derive a related decreasing spending rule when future arrivals are known in advance.

These works motivate the *decreasing-allocation policy class* defined in this paper. Under such a policy, each arrival is allocated over future periods according to the same decreasing sequence. In our model, this allocation begins one period after the arrival because the controller does not observe the current arrival before choosing the action.

The models in [10]–[12] use random arrivals from a fixed model, reveal the current harvested energy before allocation, and optimize a fixed reward that depends on the allocated power. In our setting, the arrival and cost sequences may be chosen adversarially, the controller acts before observing the current arrival and cost, the cost can depend on both the storage state and the action, and performance is measured by regret against a fixed policy.

Yu and Neely [13] study expected utility maximization under i.i.d. system states. Asgari and Neely [14] obtain an $O(\sqrt{T})$ regret–battery-capacity tradeoff for arbitrary loss sequences while retaining i.i.d. energy arrivals. Both works use objectives that depend only on the action and compare against a fixed action. Here, the arrivals may be chosen adversarially, the cost depends on the state and action, and the comparator is a fixed policy rather than a fixed action.

Lin et al. [15] study online age-of-information optimization in an energy-harvesting system with related battery dynamics. They compare their controller with an unrestricted offline optimum using a competitive ratio. Their objective depends on the energy allocation but not directly on the battery state, and the current arrival is observed before the allocation. Our controller acts before observing the current arrival, and our benchmark is a fixed policy chosen in hindsight.

c) Queueing and network optimization: Queueing control also involves states that change through arrivals and service decisions. Classical work studies throughput and queue stability [16]–[18], while stochastic network optimization studies tradeoffs between time-average objectives and queue backlog [19]. Our setting instead allows adversarial arrival sequences and convex state–action costs, and measures regret against a policy chosen in hindsight.

In many queueing models, $\alpha = 1$ because stored packets do not disappear over time. Such models are not direct examples of the $\alpha \in (0,1)$ system studied here. A queue with abandonment or expiration can have $\alpha < 1$ and is more closely related to our model. Section VII discusses the case when $\alpha \geq 1$.

Khojastepour and Sabharwal [20] study a fixed scheduler that can be written as a linear filter of past arrivals. Their scheduler minimizes average transmit power subject to a maximum-delay guarantee. In contrast, our controller updates its filter weights online in response to convex state–action costs revealed after each decision.

d) Inventory and reservoir management: Yang et al. [21] study online procurement under inventory constraints. Their controller observes the current demand and price and then chooses how much supply to purchase. Performance is measured by a competitive ratio for the total procurement cost. Our controller instead chooses how much stored resource to use before observing the current arrival and cost.

Hihat and Fermanian [22] consider inventory systems with more general state-transition functions and study policies that replenish inventory up to a target level. Their model is broader in its transition dynamics, but it has a different decision order and a different policy class. Our scalar linear dynamics allow explicit feasibility, approximation, and regret guarantees.

Classical reservoir models also use storage dynamics with random inflows and controlled releases [23], [24]. That literature mainly studies long-run performance under stochastic inflows. It provides another example of control under storage constraints, although physical reservoir models may also include capacity and overflow effects that are not part of our main model.

E. Paper Organization

Section III states the storage dynamics, information pattern, cost assumptions, feasible domain, and regret benchmark. Section IV introduces the SDAC policy class and proves its feasibility and convexity properties. It also shows how SDAC approximates decreasing-allocation policies and proportional state-feedback policies. Section V presents the feasible online controller based on action clipping and entropic OMD. Section VI proves the regret upper and lower bounds and extends the upper bound to the larger policy classes. Section VII discusses extensions to $\alpha \geq 1$, vector actions, and multiple storage states. Section VIII concludes the paper, and Appendix A contains the deferred proofs.

II. Notation

For every positive integer n, we write $[n] \triangleq \{1, \ldots, n\}$ and $[z]_+ \triangleq \max\{z, 0\}$. For $\mathbf{v} \in \mathbb{R}^n$, the i-th coordinate is denoted by v_i. We write $\mathbf{1}$ for the all-ones vector, with its dimension clear from context, and $\langle \cdot, \cdot \rangle$ for the Euclidean inner product. We use $\log$ for the natural logarithm and set $w_\tau \equiv 0$ for $\tau \leq 0$.

A fixed SDAC policy with parameter $\mathbf{m}$ is denoted by $\pi_{\mathbf{m}}$, with induced state and action trajectories $x_t(\mathbf{m})$ and $u_t(\mathbf{m})$. A feasible proportional state-feedback policy with gain $k \in [0, \alpha]$ induces $x_t(k)$ and $u_t(k)$, while a decreasing allocation sequence $\mathbf{q}$ induces $x_t(\mathbf{q})$ and $u_t(\mathbf{q})$. After the online algorithm is introduced, x_t and u_t without arguments denote the realized online trajectory.

III. Model and Problem Setup

Fix a positive integer T and a known retention coefficient $\alpha \in (0,1)$. The state $x_t \in \mathbb{R}_{\geq 0}$ is the resource level at the beginning of time t. The system starts empty:

$$x_1 = 0. \tag{1}$$

At each time $t \in [T]$, the controller chooses an action $u_t \in \mathbb{R}_{\geq 0}$ satisfying

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives an arrival $w_t \in [0,1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

Here, $1 - \alpha$ is the fraction of the stored resource lost during one period. The constraint (2) ensures that the action does not exceed the amount remaining after this loss.

Although w_t is commonly called a disturbance in the online-control literature, we call it an *arrival* because it is nonnegative. We retain the standard names Disturbance-Action Control and Simplex Disturbance-Action Control. The nonnegativity of w_t is important both for the storage interpretation and for the feasibility properties of SDAC.

At time t, the pair (w_t, c_t) may depend on the observations through time $t-1$, but not on the current action or any random choice made by the controller at time t. The controller chooses u_t before observing w_t or c_t. After the action is selected, the arrival and the entire cost function are revealed, the controller incurs $c_t(x_t, u_t)$, and the state is updated according to (3).

For every feasible trajectory,

$$0 \leq x_{t+1} \leq \alpha x_t + 1.$$

Since $x_1 = 0$, induction gives

$$0 \leq x_t \leq \frac{1 - \alpha^{t-1}}{1 - \alpha} \leq \frac{1}{1 - \alpha}, \qquad t \in [T+1].$$

Thus, the storage state is uniformly bounded. In particular, a storage device with capacity $1/(1-\alpha)$ is sufficient to realize every feasible trajectory without overflow. The state and action constraints imply that every feasible state–action pair belongs to the compact set

$$\mathcal{D}_\alpha \triangleq \left\{(x,u) : 0 \le x \le \frac{1}{1-\alpha}, 0 \le u \le \alpha x\right\}. \tag{4}$$

Each revealed cost function has the form

$$c_t : \mathbb{R}^2_{\ge 0} \to \mathbb{R}$$

and is convex in (x,u). We assume that it satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there are constants $L_x, L_u > 0$, independent of t, such that

$$|c_t(x,u) - c_t(x',u')| \le L_x|x-x'| + L_u|u-u'| \tag{5}$$

for all $(x,u),(x',u') \in \mathcal{D}_\alpha$. We also assume access, at every $(x,u) \in \mathcal{D}_\alpha$, to a cost subgradient

$$\boldsymbol{\zeta}_t(x,u) = \big(\zeta_t^x(x,u), \zeta_t^u(x,u)\big) \in \partial c_t(x,u)$$

satisfying

$$|\zeta_t^x(x,u)| \le L_x, \qquad |\zeta_t^u(x,u)| \le L_u.$$

These subgradient bounds hold, for example, when (5) extends to an open neighborhood of $\mathcal{D}_\alpha$; see [25, Thm. 9.13].

Let $\{(x_t,u_t)\}_{t=1}^T$ denote the trajectory generated by the online controller. Its cumulative cost is

$$J_T^{\mathrm{on}} \triangleq \sum_{t=1}^T c_t(x_t,u_t). \tag{6}$$

Fix a positive integer H, and let $\mathbf{m}$ range over the SDAC parameter set $\mathcal{M}_H$, defined in Section IV. For each $\mathbf{m} \in \mathcal{M}_H$, let $x_t(\mathbf{m})$ and $u_t(\mathbf{m})$ denote the trajectory generated by the fixed policy $\pi_{\mathbf{m}}$ under the same arrival sequence. Define its cumulative cost by

$$J_T(\mathbf{m}) \triangleq \sum_{t=1}^T c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big).$$

Both the online trajectory and every benchmark trajectory are required to satisfy (2) under (3). The regret against the best fixed SDAC policy is defined by

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \triangleq J_T^{\mathrm{on}} - \min_{\mathbf{m}\in\mathcal{M}_H} J_T(\mathbf{m}). \tag{7}$$

The online controller may update its SDAC parameter over time, while the benchmark in (7) is the best single fixed SDAC policy chosen in hindsight. Here, fixed means that $\mathbf{m}$ is held constant; the policy's actions still respond to past arrivals. Our objective is to satisfy (2) at every time and achieve sublinear regret.

IV. Simplex Disturbance-Action Control (SDAC)

A. From Disturbance-Action Control to SDAC

DAC combines a stabilizing state-feedback term with a finite linear filter of past disturbances. Under standard conditions, this parameterization yields convex surrogate losses and efficient online updates. In our scalar storage model, specializing the baseline feedback term to zero gives the representative form

$$u_t = \sum_{i=1}^H \theta_i w_{t-i}, \tag{8}$$

where H is a positive integer (the memory horizon) and $\boldsymbol{\theta} = (\theta_1,\dots,\theta_H)^\top \in \mathbb{R}^H$ is a linear filter.

For our constrained model, an unrestricted filter is insufficient: its action can violate (2) by exceeding αx_t or becoming negative. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which uses a simplex-constrained filter tailored to the storage dynamics. This restriction ensures that every fixed policy is feasible while preserving convex surrogate losses and tractable online updates.

B. SDAC Definition

Fix a memory horizon $H \in \mathbb{N}_{>0}$. Let

$$\mathcal{M}_H \triangleq \big\{\mathbf{m} \in \mathbb{R}^H_{\ge 0} : \ \|\mathbf{m}\|_1 \le 1\big\} \tag{9}$$

be the subprobability simplex. For any $\mathbf{m} \in \mathcal{M}_H$, define the policy $\pi_{\mathbf{m}}$ by

$$u_t(\mathbf{m}) \ \triangleq \ \sum_{i=1}^H m_i \alpha^i w_{t-i} = \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle, \tag{10}$$

where

$$\boldsymbol{\phi}_t^u \triangleq \big(\alpha w_{t-1}, \alpha^2 w_{t-2}, \dots, \alpha^H w_{t-H}\big)^\top \in \mathbb{R}^H, \tag{11}$$

The induced state trajectory satisfies

$$x_{t+1}(\mathbf{m}) = \alpha x_t(\mathbf{m}) - u_t(\mathbf{m}) + w_t, \qquad x_1(\mathbf{m}) = 0. \tag{12}$$

The SDAC policy class is

$$\Pi_H^{\mathrm{SDAC}} \triangleq \{\pi_{\mathbf{m}} : \mathbf{m} \in \mathcal{M}_H\}. \tag{13}$$

The weights α^i align the arrival filter with the retention coefficient α in (3).

We now establish the two structural properties that make SDAC useful: the induced control action always satisfies the feasibility constraint (2), and the induced cost is convex and Lipschitz in $\mathbf{m}$, so that regret minimization against the best fixed SDAC policy is an instance of online convex optimization over the simplex. Both properties follow from an explicit decomposition of the state trajectory that isolates its dependence on $\mathbf{m}$.

Lemma 1 (State decomposition under SDAC)**.** *Fix a positive integer H and $\mathbf{m} \in \mathcal{M}_H$. For $i \in \{1,\dots,H\}$, define*

$$r_{t,i} \triangleq \sum_{j=1}^i \alpha^{j-1} w_{t-j}, \qquad r_t \triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j},$$

where the second sum is empty when $t = 1$. Then, for every $t \in [T]$,

$$x_t(\mathbf{m}) = \sum_{i=1}^{H} m_i r_{t,i} + \big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t. \tag{14}$$

Define the quantity

$$\boldsymbol{\phi}_t^x \triangleq \big(r_{t,1} - r_t, \dots, r_{t,H} - r_t\big)^\top.$$

Then

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle.$$

Moreover,

$$0 \le r_{t,i} \le r_t \le \frac{1}{1-\alpha},$$

and $\mathbf{m} \mapsto x_t(\mathbf{m})$ is affine.

Proof. The auxiliary quantities satisfy

$$r_{t+1} = \alpha r_t + w_t, \qquad r_1 = 0, \tag{15}$$

and, for every $i \in \{1, \dots, H\}$,

$$\begin{aligned} \alpha r_{t,i} - \alpha^i w_{t-i} &= \sum_{j=1}^{i-1} \alpha^j w_{t-j} \\ &= r_{t+1,i} - w_t. \end{aligned} \tag{16}$$

For $i = 1$, the sum in the first line is empty.

At $t = 1$, since $w_\tau \equiv 0$ for $\tau \le 0$,

$$\begin{aligned} r_{1,i} &= \sum_{j=1}^{i} \alpha^{j-1} w_{1-j} = 0, \\ r_1 &= 0, \\ x_1(\mathbf{m}) &= 0, \end{aligned}$$

so (14) holds. Suppose it holds at time t. Using (12) and (10),

$$\begin{aligned} x_{t+1}(\mathbf{m}) = \alpha &\left(\sum_{i=1}^{H} m_i r_{t,i} + \big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t \right) \\ &- \sum_{i=1}^{H} m_i \alpha^i w_{t-i} + w_t \\ =& \sum_{i=1}^{H} m_i \big(\alpha r_{t,i} - \alpha^i w_{t-i}\big) \\ &+ \alpha \big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t + w_t. \end{aligned} \tag{17}$$

Substituting (15) and (16) into (17),

$$\begin{aligned} x_{t+1}(\mathbf{m}) =& \sum_{i=1}^{H} m_i \big(r_{t+1,i} - w_t\big) \\ &+ \big(1 - \mathbf{1}^\top \mathbf{m}\big)(r_{t+1} - w_t) + w_t \\ =& \sum_{i=1}^{H} m_i r_{t+1,i} + \big(1 - \mathbf{1}^\top \mathbf{m}\big) r_{t+1}, \end{aligned} \tag{18}$$

because the total coefficient of w_t in the first line is $-\mathbf{1}^\top \mathbf{m} - \big(1 - \mathbf{1}^\top \mathbf{m}\big) + 1 = 0$. This proves (14) by induction.

All summands defining $r_{t,i}$ and r_t are nonnegative. Since $w_\tau = 0$ for $\tau \le 0$, $r_{t,i}$ is a truncation of r_t; hence

$$0 \le r_{t,i} \le r_t \le \sum_{j=1}^{\infty} \alpha^{j-1} = \frac{1}{1-\alpha}.$$

Finally,

$$x_t(\mathbf{m}) = r_t + \sum_{i=1}^{H} m_i \big(r_{t,i} - r_t\big) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle,$$

which is affine in $\mathbf{m}$. □

The first consequence of this decomposition is that every fixed policy is feasible.

Lemma 2 (Feasibility of SDAC policies)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the trajectory induced by* (10) *and* (12) *satisfies the action constraint* (2)*; explicitly,*

$$0 \le u_t(\mathbf{m}) \le \alpha x_t(\mathbf{m}) \qquad \textit{for all } t \in [T]. \tag{19}$$

Proof. Because m_i, α, and w_{t-i} are nonnegative,

$$u_t(\mathbf{m}) = \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \ge 0.$$

For every $i \in \{1, \dots, H\}$,

$$\alpha r_{t,i} = \sum_{j=1}^{i} \alpha^j w_{t-j} \ge \alpha^i w_{t-i}. \tag{20}$$

Because $\mathbf{m} \in \mathcal{M}_H$ and $r_t \ge 0$ by Lemma 1,

$$\begin{aligned} \alpha x_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \alpha r_{t,i} + \alpha \big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t \\ &\ge \sum_{i=1}^{H} m_i \alpha r_{t,i} \\ &\ge \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = u_t(\mathbf{m}), \end{aligned} \tag{21}$$

where the second inequality uses (20). □

For fixed arrivals, both coordinates of the SDAC state–action pair are affine in $\mathbf{m}$. The following lemma records the resulting convexity and the constants used by the online update.

Lemma 3 (Convexity and Lipschitz continuity of surrogate cost)**.** *Define*

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H,$$

and

$$G_\alpha \triangleq \alpha \left(\frac{L_x}{1-\alpha} + L_u \right). \tag{22}$$

Then ℓ_t is convex on $\mathcal{M}_H$, and for all $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| \le G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.$$

Moreover, at every $\mathbf{m} \in \mathcal{M}_H$, one can select $\mathbf{g}_t \in \partial \ell_t(\mathbf{m})$ such that $\|\mathbf{g}_t\|_\infty \le G_\alpha$.

Proof. Lemma 1 and (10) give

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m}\rangle, \qquad u_t(\mathbf{m}) = \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle. \tag{23}$$

For every $i \in \{1, \dots, H\}$, the definitions of $r_{t,i}$ and r_t give

$$\begin{aligned}(\boldsymbol{\phi}_t^x)_i &= r_{t,i} - r_t \\ &= -\sum_{j=i+1}^{t-1} \alpha^{j-1} w_{t-j},\end{aligned} \tag{24}$$

where the sum is empty when $i \geq t-1$. Hence

$$\begin{aligned}\left|(\boldsymbol{\phi}_t^x)_i\right| &\leq \sum_{j=i+1}^{\infty} \alpha^{j-1} = \frac{\alpha^i}{1-\alpha}, \\ 0 \leq (\boldsymbol{\phi}_t^u)_i &= \alpha^i w_{t-i} \leq \alpha^i.\end{aligned} \tag{25}$$

Taking the largest coordinate, attained in these upper bounds at $i = 1$, yields

$$\|\boldsymbol{\phi}_t^x\|_\infty \leq \frac{\alpha}{1-\alpha}, \qquad \|\boldsymbol{\phi}_t^u\|_\infty \leq \alpha. \tag{26}$$

Since (23) is affine in $\mathbf{m}$ and c_t is convex, ℓ_t is convex.

By (5) and (23),

$$\begin{aligned}|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| &\leq L_x\left|\langle \boldsymbol{\phi}_t^x, \mathbf{m}-\mathbf{m}'\rangle\right| + L_u\left|\langle \boldsymbol{\phi}_t^u, \mathbf{m}-\mathbf{m}'\rangle\right| \\ &\leq \left(L_x\|\boldsymbol{\phi}_t^x\|_\infty + L_u\|\boldsymbol{\phi}_t^u\|_\infty\right)\|\mathbf{m}-\mathbf{m}'\|_1 \\ &\leq \alpha\left(\frac{L_x}{1-\alpha} + L_u\right)\|\mathbf{m}-\mathbf{m}'\|_1 \\ &= G_\alpha\|\mathbf{m}-\mathbf{m}'\|_1.\end{aligned} \tag{27}$$

Let (ζ_t^x, ζ_t^u) denote the components of the bounded subgradient $\boldsymbol{\zeta}_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big)$ from Section III, and set

$$\mathbf{g}_t \triangleq \zeta_t^x\, \boldsymbol{\phi}_t^x + \zeta_t^u\, \boldsymbol{\phi}_t^u. \tag{28}$$

For every $\mathbf{m}' \in \mathcal{M}_H$, the subgradient inequality and (23) yield

$$\begin{aligned}\ell_t(\mathbf{m}') &\geq \ell_t(\mathbf{m}) + \zeta_t^x\langle \boldsymbol{\phi}_t^x, \mathbf{m}'-\mathbf{m}\rangle + \zeta_t^u\langle \boldsymbol{\phi}_t^u, \mathbf{m}'-\mathbf{m}\rangle \\ &= \ell_t(\mathbf{m}) + \langle \mathbf{g}_t, \mathbf{m}'-\mathbf{m}\rangle.\end{aligned}$$

Thus $\mathbf{g}_t \in \partial\ell_t(\mathbf{m})$. For each $i \in \{1, \dots, H\}$, the exact coordinate decay in (25) gives

$$\begin{aligned}|g_{t,i}| &\leq |\zeta_t^x|\left|(\boldsymbol{\phi}_t^x)_i\right| + |\zeta_t^u|(\boldsymbol{\phi}_t^u)_i \\ &\leq \alpha^i\left(\frac{L_x}{1-\alpha} + L_u\right) \\ &\leq \alpha\left(\frac{L_x}{1-\alpha} + L_u\right) = G_\alpha.\end{aligned} \tag{29}$$

Taking the maximum over i proves

$$\|\mathbf{g}_t\|_\infty \leq G_\alpha. \tag{30}$$

□

Regret minimization against the best fixed SDAC policy thus becomes online convex optimization over the simplex.

C. Expressiveness: Decreasing-Allocation Policies

Energy-harvesting policies for Bernoulli recharges can be described by the sequence of expenditures following each recharge [11], [12]. In the corresponding decreasing-allocation policy class, each unit arriving at time s is assigned the same decreasing allocation sequence, beginning at time $s+1$ to respect the information pattern. The sequence may have an arbitrary decreasing shape; geometric sequences induce proportional state-feedback policies.

Define the set of feasible decreasing allocation sequences

$$\mathcal{Q}_\alpha^{\downarrow} \triangleq \left\{\mathbf{q} = (q_i)_{i\geq 1} : q_1 \geq q_2 \geq \cdots \geq 0, \quad \sum_{i=1}^{\infty}\frac{q_i}{\alpha^i} \leq 1\right\}. \tag{31}$$

For $\mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}$, define

$$\begin{aligned}u_t(\mathbf{q}) &\triangleq \sum_{i=1}^{t-1} q_i w_{t-i}, \\ x_{t+1}(\mathbf{q}) &= \alpha x_t(\mathbf{q}) - u_t(\mathbf{q}) + w_t, \qquad x_1(\mathbf{q}) = 0.\end{aligned} \tag{32}$$

Here $q_i w_s$ is the amount from arrival w_s allocated i periods later. The weighted budget in (31) accounts for leakage: reserving q_i units for expenditure i periods after an arrival requires q_i/α^i units at the arrival time. Let

$$\Pi_\alpha^{\text{alloc}} \triangleq \{\pi_{\mathbf{q}} : \mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}\}.$$

Lemma 4 (Finite-memory approximation of decreasing-allocation policies)**.** *Every policy in $\Pi_\alpha^{\text{alloc}}$ is feasible. Fix $\mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}$ and $H \geq 1$, and define*

$$m_{H,i}(\mathbf{q}) \triangleq \frac{q_i}{\alpha^i}, \qquad i \in [H]. \tag{33}$$

Then $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. With

$$\mathrm{Tail}_H(\mathbf{q}) \triangleq \sum_{i=H+1}^{\infty} q_i,$$

the corresponding trajectories satisfy, for every arrival sequence in $[0,1]$,

$$0 \leq u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big) \leq \mathrm{Tail}_H(\mathbf{q}), \tag{34}$$

$$0 \leq x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}) \leq \frac{\mathrm{Tail}_H(\mathbf{q})}{1-\alpha}. \tag{35}$$

If each c_t satisfies (5)*, then*

$$\begin{aligned}&\left|\sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big) - \sum_{t=1}^{T} c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\right| \\ &\quad\leq \left(L_u + \frac{L_x}{1-\alpha}\right)\mathrm{Tail}_H(\mathbf{q})T \\ &\quad\leq \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H+1}T.\end{aligned} \tag{36}$$

In particular, fix $k \in [0, \alpha]$, set $\lambda_k \triangleq \alpha - k$, and define

$$q_i(k) = k\lambda_k^{i-1}, \qquad i \geq 1, \tag{37}$$

where $\lambda_k^0 \triangleq 1$. Then $\mathbf{q}(k) \in \mathcal{Q}_\alpha^{\downarrow}$ and $\pi_{\mathbf{q}(k)}$ is exactly the feasible proportional state-feedback policy $u_t(k) = kx_t(k)$. Its SDAC truncation is

$$\left[\mathbf{m}_H^{\text{prop}}(k)\right]_i \triangleq m_{H,i}\big(\mathbf{q}(k)\big) = \frac{k\lambda_k^{i-1}}{\alpha^i}, \qquad i \in [H],$$

and its tail is

$$\mathrm{Tail}_H\big(\mathbf{q}(k)\big) = \frac{k\lambda_k^H}{1-\lambda_k}.$$

Therefore, (36) *specializes exactly to*

$$\left|\sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}_H^{\mathrm{prop}}(k)), u_t(\mathbf{m}_H^{\mathrm{prop}}(k))\big) - \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big)\right| \le \left(L_u + \frac{L_x}{1-\alpha}\right)\frac{k\lambda_k^H}{1-\lambda_k}\,T. \tag{38}$$

The proof is deferred to Appendix A.

$\Pi_\alpha^{\mathrm{alloc}}$ contains every feasible proportional state-feedback policy and also permits nongeometric allocation sequences. Together with Theorem 1, the tail bound in (36) yields no regret against $\Pi_\alpha^{\mathrm{alloc}}$ and, consequently, against feasible proportional state-feedback policies.

V. Online Algorithm

We now present our online algorithm, which updates SDAC parameters by OMD with unnormalized negentropy (equivalently, the exponentiated gradient method) and a fixed step size $\eta > 0$ over the subprobability simplex $\mathcal{M}_H \subset \mathbb{R}^H$ [26]. Let

$$\Psi(\mathbf{m}) \triangleq \sum_{i=1}^{H}\big(m_i \log m_i - m_i\big), \qquad \mathbf{m} \in \mathbb{R}^H_{\ge 0}, \tag{39}$$

denote the unnormalized negentropy (with the convention $0\log 0 \triangleq 0$). Its Bregman divergence is the generalized Kullback–Leibler divergence

$$D_\Psi(\mathbf{m}\|\mathbf{v}) \triangleq \sum_{i=1}^{H}\Big(m_i \log \tfrac{m_i}{v_i} - m_i + v_i\Big), \tag{40}$$

defined for

$$\mathbf{m} \in \mathbb{R}^H_{\ge 0}, \qquad \mathbf{v} \in \mathbb{R}^H_{>0}.$$

For each time $t \in [T]$, let $\mathbf{m}_t \in \mathcal{M}_H$ be the online parameter. We initialize at

$$\mathbf{m}_1 \triangleq \frac{1}{H+1}\mathbf{1} \tag{41}$$

so that $\sup_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\mathbf{m}_1) \le \log(H+1)$ (see the proof of Theorem 1).

a) Control-action selection: The controller first forms the nominal SDAC action

$$\widehat{u}_t \triangleq \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \tag{42}$$

and clips it to the admissible interval in (2):

$$u_t \triangleq \min\{\widehat{u}_t,\, \alpha x_t\}. \tag{43}$$

The state then evolves according to (3). Because $\widehat{u}_t \ge 0$, the clipping rule and induction on (3), starting from (1) with $w_t \in [0,1]$, yield $x_t \ge 0$ for all $t \in [T+1]$ and show that the online trajectory satisfies (2).

The scalar correction in (43) also has an exact Bregman-projection equivalent.

Lemma 5 (Action clipping as a Bregman projection)**.** *At time t, define the parameter set compatible with the available state by*

$$\mathcal{F}_t(x_t) \triangleq \{\mathbf{m} \in \mathcal{M}_H : \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle \le \alpha x_t\}. \tag{44}$$

For the negentropy Ψ *in* (39)*, let*

$$\mathbf{m}_t^{\mathrm{exec}} \in \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{F}_t(x_t)} D_\Psi(\mathbf{m}\|\mathbf{m}_t). \tag{45}$$

Then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\} = u_t. \tag{46}$$

Thus the projection formulation and the scalar clipping rule induce exactly the same executed action.

Proof. The set $\mathcal{F}_t(x_t)$ is nonempty because it contains $\mathbf{0}$, and it is compact and convex. Since $\mathbf{m}_t$ is strictly positive coordinatewise, $D_\Psi(\cdot\|\mathbf{m}_t)$ is strictly convex on $\mathcal{M}_H$ and is minimized uniquely at $\mathbf{m}_t$ over any feasible set containing $\mathbf{m}_t$. Consequently, if

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle \le \alpha x_t,$$

then $\mathbf{m}_t^{\mathrm{exec}} = \mathbf{m}_t$, which proves (46) in the feasible case.

Suppose instead that $\mathbf{m}_t$ is infeasible. We claim that the action constraint is active at $\mathbf{m}_t^{\mathrm{exec}}$. If it were slack, then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle < \alpha x_t < \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle. \tag{47}$$

For sufficiently small $\lambda \in (0,1)$, the point

$$\mathbf{m}_\lambda \triangleq (1-\lambda)\mathbf{m}_t^{\mathrm{exec}} + \lambda\mathbf{m}_t$$

would still belong to $\mathcal{F}_t(x_t)$. Strict convexity and $D_\Psi(\mathbf{m}_t\|\mathbf{m}_t) = 0$ would then give

$$\begin{aligned} D_\Psi(\mathbf{m}_\lambda\|\mathbf{m}_t) &< (1-\lambda)D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t) \\ &< D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t), \end{aligned}$$

contradicting (45). Hence

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \alpha x_t$$

in the infeasible case, which completes the proof. □

The implemented rule remains the scalar minimum (43); the full H-dimensional vector $\mathbf{m}_t^{\mathrm{exec}}$ is never calculated because only its scalar induced action is executed. Forming the feature vectors and their inner products, followed by the OMD update below, takes $O(H)$ time and $O(H)$ memory per round. In particular, the next unconstrained OMD update uses the internal iterate $\mathbf{m}_t$, not $\mathbf{m}_t^{\mathrm{exec}}$. Therefore the internal iterates remain on the fixed domain $\mathcal{M}_H$, and the analysis does not acquire a time-varying comparator set. The direct implementation and its projection interpretation produce the same actions, states, and internal OMD iterates.

b) Surrogate cost and OMD update: After (w_t, c_t) is revealed, select a subgradient of the convex surrogate loss defined in Lemma 3, $\mathbf{g}_t \in \partial\ell_t(\mathbf{m}_t)$ satisfying $\|\mathbf{g}_t\|_\infty \le G_\alpha$ as in (22), whose existence and construction are given in that lemma. This step uses the revealed cost function and the full-information subgradient oracle from Section III. We define the OMD update

$$\mathbf{m}_{t+1} \triangleq \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H}\Big\{\eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\Big\}. \tag{48}$$

c) Closed-form solution: Although (48) is a constrained optimization problem, it is easy to solve analytically because Ψ is separable and the objective is linear in $\mathbf{g}_t$. As shown in Lemma 7, the minimizer in (48) is exactly the result of two elementary steps. First a multiplicative step,

$$\widetilde{m}_{t+1,i} \triangleq m_{t,i} \exp\bigl(-\eta g_{t,i}\bigr), \qquad i = 1, \ldots, H, \tag{49}$$

where $g_{t,i}$ is the i-th coordinate of $\mathbf{g}_t$. This is followed by the Bregman projection onto $\mathcal{M}_H$,

$$\mathbf{m}_{t+1} \triangleq \mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}_{t+1}) = \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}, \tag{50}$$

where the projection operator is defined and derived in Lemma 6; the scaling is nontrivial only when $\|\widetilde{\mathbf{m}}_{t+1}\|_1 > 1$.

VI. Regret Analysis

The following theorem states that the online controller achieves sublinear regret against every fixed SDAC policy $\pi_{\mathbf{m}}$, where $\mathbf{m}$ is any parameter in $\mathcal{M}_H$. The regret proof has two parts: a standard online convex optimization bound for the surrogate costs ℓ_t, and a stability bound showing that the online trajectory stays close to the SDAC policy trajectory driven by the same parameters. Some technical lemmas required for the proof are deferred to Appendix A.

Theorem 1 (Regret bound)**.** *Suppose the cost functions satisfy the regularity and bounded-subgradient assumptions of Section III. Then the OMD controller with action clipping from Section V, initialized as in* (41) *and using step size* $\eta > 0$*, satisfies, for the regret defined in* (7),

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha, \tag{51}$$

where

$$\Gamma_\alpha \triangleq \frac{G_\alpha^2}{2} + \frac{\alpha G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}, \tag{52}$$

and $G_\alpha = \alpha\bigl(\frac{L_x}{1-\alpha} + L_u\bigr)$ *as in* (22)*. In particular, the choice*

$$\eta^* = \sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}$$

gives

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le 2\sqrt{\Gamma_\alpha T \log(H+1)}.$$

For fixed $L_x, L_u > 0$*, the leading factor obeys*

$$\sqrt{\Gamma_\alpha} = \Theta(\alpha) \quad \textit{as } \alpha \downarrow 0,$$

and

$$\sqrt{\Gamma_\alpha} = \Theta\Bigl((1-\alpha)^{-3/2}\Bigr) \quad \textit{as } \alpha \uparrow 1.$$

The next result shows that the $\sqrt{T}$ dependence is unavoidable and quantifies the dependence on $1-\alpha$ that any causal feasible controller must incur. Its proof is deferred to Appendix A.

Theorem 2 (Minimax SDAC regret lower bound)**.** *Fix* $H \ge 1$ *and* $T \ge 2$*. For every causal feasible online controller* $\mathcal{A}$*, possibly randomized, there exists a deterministic oblivious sequence of arrivals and costs satisfying the assumptions of Section III such that*

$$\mathbb{E}_{\mathcal{A}}\bigl[\mathrm{Reg}_T^{\mathrm{SDAC}}\bigr] \ge \frac{1}{2\sqrt{2}} \sqrt{\sum_{t=2}^{T} \beta_t^2}, \tag{53}$$

where

$$\beta_t \triangleq \alpha\left(L_u + \frac{L_x(1-\alpha^{t-2})}{1-\alpha}\right). \tag{54}$$

The expectation is only over the possible internal randomization of $\mathcal{A}$*; the adversarial sequence whose existence is asserted is deterministic and fixed before the interaction.*

Corollary 1 (Lower bound for sufficiently long horizons)**.** *Define*

$$t_{\mathrm{mix}}(\alpha) \triangleq \left\lceil \frac{\log 2}{\log(1/\alpha)} \right\rceil. \tag{55}$$

If $T \ge 2t_{\mathrm{mix}}(\alpha) + 2$*, then every causal feasible controller* $\mathcal{A}$ *admits a deterministic oblivious sequence for which*

$$\mathbb{E}_{\mathcal{A}}\bigl[\mathrm{Reg}_T^{\mathrm{SDAC}}\bigr] \ge \frac{G_\alpha}{8}\sqrt{T}, \tag{56}$$

where $G_\alpha = \alpha(L_x/(1-\alpha) + L_u)$ *as in* (22)*.*

The limit $\alpha \uparrow 1$ is more challenging because stored arrivals persist longer. In this limit, $t_{\mathrm{mix}}(\alpha) = \Theta((1-\alpha)^{-1})$, and Corollary 1 gives

$$\mathbb{E}_{\mathcal{A}}\bigl[\mathrm{Reg}_T^{\mathrm{SDAC}}\bigr] = \Omega\left(\frac{\sqrt{T}}{1-\alpha}\right) \tag{57}$$

whenever $T \ge 2t_{\mathrm{mix}}(\alpha) + 2$. The upper bound in Theorem 1 is $O((1-\alpha)^{-3/2}\sqrt{T\log(H+1)})$, so a factor $(1-\alpha)^{-1/2}$ between the two bounds remains open, apart from the memory-dependent logarithm.

The condition $T \ge 2t_{\mathrm{mix}}(\alpha) + 2$ is essential for interpreting the large-α asymptotic. If T is fixed while $\alpha \uparrow 1$, then

$$\frac{1-\alpha^{t-2}}{1-\alpha} \longrightarrow t-2,$$

and the right-hand side of (53) converges to the finite quantity

$$\frac{1}{2\sqrt{2}} \sqrt{\sum_{t=2}^{T} \bigl(L_u + L_x(t-2)\bigr)^2}.$$

Thus the exact finite-horizon formula, rather than the long-horizon scaling in (57), applies in this regime.

The large-α dependence improves the coarser analysis that ignores the exact coordinate decay in (25) and the nonexpansiveness of $z \mapsto [z]_+$ used in (94). That analysis gives a leading factor of order $(1-\alpha)^{-2}$ as $\alpha \uparrow 1$. The sharper rate follows from the gradient and stability estimates.

Theorem 1 and Lemma 4 give the following comparator guarantee.

Corollary 2 (No-regret against decreasing-allocation policies)**.** *Set*

$$H_T \triangleq \max\left\{1, \left\lceil \frac{\log T}{\log(1/\alpha)} \right\rceil\right\},$$

and run the OMD controller with action clipping, memory horizon $H = H_T$, and the step size η^* of Theorem 1. For $\mathbf{q} \in \mathcal{Q}_\alpha^\downarrow$, let

$$J_T(\mathbf{q}) \triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)$$

and

$$\mathrm{Reg}_T^{\mathrm{alloc}} \triangleq J_T^{\mathrm{on}} - \inf_{\mathbf{q}\in\mathcal{Q}_\alpha^\downarrow} J_T(\mathbf{q}).$$

Then

$$\mathrm{Reg}_T^{\mathrm{alloc}} \le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T. \tag{58}$$

Here Γ_α is defined in (52)*. Since $\alpha^{H_T} \le 1/T$ and $\log(H_T + 1) = O(\log\log T)$ as $T \to \infty$, the approximation term is $O(1)$. Consequently,*

$$\mathrm{Reg}_T^{\mathrm{alloc}} = O\Big(\sqrt{T\log\log T}\Big) = o(T).$$

Proof. Fix $\mathbf{q} \in \mathcal{Q}_\alpha^\downarrow$ and use the SDAC comparator $\mathbf{m}_{H_T}(\mathbf{q})$ from Lemma 4. Theorem 1 and (36) give

$$\begin{aligned} J_T^{\mathrm{on}} - J_T(\mathbf{q}) &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\mathrm{Tail}_{H_T}(\mathbf{q})T \\ &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T. \end{aligned}$$

The bound is uniform in $\mathbf{q}$, so taking the infimum comparator cost proves (58). Finally, $H_T \ge \log T/\log(1/\alpha)$ implies $\alpha^{H_T} \le 1/T$, while $H_T = \Theta(\log T)$ implies $\log(H_T + 1) = O(\log\log T)$. □

Because geometric allocation sequences induce proportional state feedback, the same controller also satisfies a gain-dependent bound for that policy family.

Corollary 3 (No-regret against proportional state-feedback policies)**.** *Run the controller of Corollary 2. For $k \in [0, \alpha]$, let $J_T(k) \triangleq \sum_{t=1}^T c_t\big(x_t(k), u_t(k)\big)$, where $u_t(k) = kx_t(k)$, and define*

$$\mathrm{Reg}_T^{\mathrm{prop}} \triangleq J_T^{\mathrm{on}} - \min_{k\in[0,\alpha]} J_T(k).$$

For every $k \in [0, \alpha]$, with $\lambda_k = \alpha - k$,

$$J_T^{\mathrm{on}} - J_T(k) \le 2\sqrt{\Gamma_\alpha T\log(H_T+1)} + \left(L_u + \frac{L_x}{1-\alpha}\right)\frac{k\lambda_k^{H_T}}{1-\lambda_k}T. \tag{59}$$

Moreover,

$$\begin{aligned} \mathrm{Reg}_T^{\mathrm{prop}} &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T \\ &= O\Big(\sqrt{T\log\log T}\Big) = o(T). \end{aligned}$$

Proof. For fixed k, Theorem 1 applied to $\mathbf{m}_{H_T}^{\mathrm{prop}}(k)$, followed by (38), gives (59). The geometric allocation sequence $\mathbf{q}(k)$ belongs to $\mathcal{Q}_\alpha^\downarrow$, so $\mathrm{Tail}_{H_T}(\mathbf{q}(k)) \le \alpha^{H_T+1}$ by Lemma 4. This makes the bound uniform in k; minimizing over k proves the stated finite-time and asymptotic guarantees. □

VII. Extensions

We briefly discuss three possible extensions. Establishing their full regret bounds is left for future work.

a) Retention $\alpha \ge 1$.: For $\alpha \ge 1$, choose a fixed baseline drain kx_t such that $\bar\alpha \triangleq \alpha - k \in (0, 1)$, and write $u_t = kx_t + v_t$. The residual dynamics become $x_{t+1} = \bar\alpha x_t - v_t + w_t$. An SDAC filter for v_t should therefore use the powers $\bar\alpha^i$, not α^i. If $0 \le v_t \le \bar\alpha x_t$, then $0 \le u_t \le (k + \bar\alpha)x_t = \alpha x_t$, so (2) holds. This reduction shows that the generalization to a fixed, given k is straightforward: the present analysis carries over with transformed cost and Lipschitz constants. We therefore omit the details, which would amount to extra writing without additional novelty. The choice of k itself remains an interesting avenue for future work.

b) Vector action / multi-task allocation.: Consider a single storage state x_t with a vector action $\mathbf{u}_t \in \mathbb{R}^n_{\ge 0}$ that allocates resources across n tasks, subject to $\mathbf{1}^\top\mathbf{u}_t \le \alpha x_t$, so that the total drain from storage is the sum of the task allocations. The dynamics remain $x_{t+1} = \alpha x_t - \mathbf{1}^\top\mathbf{u}_t + w_t$, and the cost $c_t(x_t, \mathbf{u}_t)$ is convex in $(x, \mathbf{u})$. The fixed-policy SDAC construction lifts by replacing the filter vector $\mathbf{m} \in \mathcal{M}_H$ with a matrix $\mathbf{M} \in \mathbb{R}^{n\times H}_{\ge 0}$ whose entries $M_{a,i}$ sum to at most one, with each row a corresponding to a task and each column i to a memory lag. OMD can then run over this set. If the nominal total allocation exceeds αx_t, the action vector is scaled by a nonnegative factor so that $\mathbf{1}^\top\mathbf{u}_t = \alpha x_t$.

c) Multiple storages with diagonal dynamics.: Suppose the state is $\mathbf{x}_t \in \mathbb{R}^d_{\ge 0}$ and evolves as $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t - \mathbf{u}_t + \mathbf{w}_t$ with $\mathbf{A} = \mathrm{diag}(\alpha_1, \ldots, \alpha_d)$, $\alpha_i \in (0, 1)$, arrivals $\mathbf{w}_t \in [0, 1]^d$, and per-coordinate constraints $0 \le u_{t,i} \le \alpha_i x_{t,i}$. The dynamics and feasible set then decouple by coordinate, and the natural SDAC class is a product of simplices. For a jointly convex cost, OMD operates jointly on this product set; the optimization separates by coordinate only when the cost is coordinate-separable. Cross-coupled storage systems (non-diagonal $\mathbf{A}$) require additional analysis.

These constructions retain the basic simplex/OMD structure, but their complete feasibility and regret analyses require separate treatment.

VIII. Conclusion

We studied adversarial online control of a storage system under a state-dependent action constraint. We constructed the SDAC so that every policy trajectory is feasible, and an online algorithm with $O\Big(\sqrt{T\log(H+1)}\Big)$ regret against the best fixed SDAC policy. Once T is at least on the order of $(1-\alpha)^{-1}$, the minimax lower bound gives $\Omega((1-\alpha)^{-1}\sqrt{T})$ regret as $\alpha \uparrow 1$; hence a factor $(1-\alpha)^{-1/2}$ between the upper and lower bounds remains open. The SDAC approximation results further give $O\big(\sqrt{T\log\log T}\big)$ regret against the best decreasing-allocation policy when $H = \Theta(\log T)$. The model applies directly to energy-harvesting battery management and related settings in which storage limits must be respected in real time.

Appendix A
Deferred Proofs

This appendix proves the minimax lower bound from Section VI and the expressiveness lemma from Section IV, together with the OMD and stability lemmas leading up to the upper regret bound.

We first prove the lower bound. Following the standard random-sign mechanism for online linear-optimization lower bounds [26], the construction embeds a two-comparator online linear problem into the two endpoint SDAC policies $\mathbf{0}$ and $\mathbf{e}_1$.

Proof of Theorem 2. Fix a causal feasible controller $\mathcal{A}$. Let $\xi_1, \ldots, \xi_T$ be independent Rademacher random variables, sampled before the interaction and independently of the internal randomization of $\mathcal{A}$. Consider the oblivious random construction

$$w_t = 1, \qquad c_t(x,u) = C_\alpha + \xi_t(L_x x - L_u u), \tag{60}$$

where

$$C_\alpha \triangleq \frac{L_x + \alpha L_u}{1-\alpha}. \tag{61}$$

Each c_t is affine and hence convex. For any two points in $\mathcal{D}_\alpha$,

$$\begin{aligned} |c_t(x,u) - c_t(x',u')| &= |L_x(x-x') - L_u(u-u')| \\ &\le L_x|x-x'| + L_u|u-u'|. \end{aligned} \tag{62}$$

The constant gradient $\xi_t(L_x, -L_u)$ also obeys the bounded-subgradient assumption. Moreover, for $(x,u) \in \mathcal{D}_\alpha$,

$$\begin{aligned} |L_x x - L_u u| &\le L_x x + L_u u \\ &\le \frac{L_x + \alpha L_u}{1-\alpha} = C_\alpha. \end{aligned} \tag{63}$$

Thus the offset even makes the costs nonnegative on the feasible domain, although it will cancel from regret.

Because $H \ge 1$, both

$$\mathbf{m}^{(0)} \triangleq \mathbf{0}, \qquad \mathbf{m}^{(1)} \triangleq \mathbf{e}_1 \tag{64}$$

belong to $\mathcal{M}_H$, where $\mathbf{e}_1$ is the first standard basis vector in $\mathbb{R}^H$. Under the arrivals in (60), the first endpoint has

$$u_t(\mathbf{m}^{(0)}) = 0, \qquad x_t(\mathbf{m}^{(0)}) = \rho_t \triangleq \frac{1-\alpha^{t-1}}{1-\alpha}. \tag{65}$$

Indeed, this follows by unrolling $x_{t+1} = \alpha x_t + 1$ from $x_1 = 0$. At the second endpoint,

$$u_1(\mathbf{m}^{(1)}) = 0, \qquad u_t(\mathbf{m}^{(1)}) = \alpha, \quad t \ge 2. \tag{66}$$

The state satisfies $x_2(\mathbf{m}^{(1)}) = 1$. If it equals one at a time $t \ge 2$, then

$$x_{t+1}(\mathbf{m}^{(1)}) = \alpha - \alpha + 1 = 1.$$

Consequently,

$$x_t(\mathbf{m}^{(1)}) = 1, \qquad t \ge 2. \tag{67}$$

For $t \ge 2$, the state difference between the endpoints is

$$\begin{aligned} \Delta_t^{\text{end}} &\triangleq x_t(\mathbf{m}^{(0)}) - x_t(\mathbf{m}^{(1)}) \\ &= \rho_t - 1 \\ &= \frac{\alpha(1-\alpha^{t-2})}{1-\alpha}. \end{aligned} \tag{68}$$

Remove the common offset by defining the centered online and endpoint costs

$$\begin{aligned} J_{T,\text{ctr}}^{\text{on}} &\triangleq \sum_{t=1}^{T} \xi_t(L_x x_t - L_u u_t), \\ J_{T,\text{ctr}}^{(j)} &\triangleq \sum_{t=1}^{T} \xi_t \left(L_x x_t(\mathbf{m}^{(j)}) - L_u u_t(\mathbf{m}^{(j)})\right), \quad j \in \{0,1\}. \end{aligned} \tag{69}$$

At time t, the pair (x_t, u_t) is determined by the past observations and the controller randomization used before c_t is revealed. It is therefore independent of the fresh sign ξ_t. Conditional mean zero gives

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[\xi_t(L_x x_t - L_u u_t) \,|\, x_t, u_t] &= 0, \\ \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\left[J_{T,\text{ctr}}^{\text{on}}\right] &= 0. \end{aligned} \tag{70}$$

Likewise, the two fixed endpoint trajectories do not depend on the signs, so

$$\mathbb{E}_{\boldsymbol{\xi}}\left[J_{T,\text{ctr}}^{(0)}\right] = \mathbb{E}_{\boldsymbol{\xi}}\left[J_{T,\text{ctr}}^{(1)}\right] = 0. \tag{71}$$

Since the two endpoints agree at $t=1$, equations (65), (66), and (68) give

$$\begin{aligned} J_{T,\text{ctr}}^{(0)} - J_{T,\text{ctr}}^{(1)} &= \sum_{t=2}^{T} \xi_t(L_x \Delta_t^{\text{end}} + \alpha L_u) \\ &= \sum_{t=2}^{T} \xi_t \beta_t, \end{aligned} \tag{72}$$

with β_t as in (54).

Minimizing over all of $\mathcal{M}_H$ can only decrease the comparator cost. After canceling the common term TC_α, the regret therefore satisfies pointwise

$$\text{Reg}_T^{\text{SDAC}} \ge J_{T,\text{ctr}}^{\text{on}} - \min\left\{J_{T,\text{ctr}}^{(0)}, J_{T,\text{ctr}}^{(1)}\right\}. \tag{73}$$

Using $\min\{a,b\} = (a+b-|a-b|)/2$ together with (70) and (71), we obtain

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\left[\text{Reg}_T^{\text{SDAC}}\right] &\ge \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left[\left|J_{T,\text{ctr}}^{(0)} - J_{T,\text{ctr}}^{(1)}\right|\right] \\ &= \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left[\left|\sum_{t=2}^{T} \xi_t \beta_t\right|\right]. \end{aligned} \tag{74}$$

Khintchine's inequality for Rademacher sums implies

$$\mathbb{E}_{\boldsymbol{\xi}}\left[\left|\sum_{t=2}^{T} \xi_t \beta_t\right|\right] \ge \frac{1}{\sqrt{2}}\left(\sum_{t=2}^{T} \beta_t^2\right)^{1/2}. \tag{75}$$

Substitution of (54) into (74)–(75) proves (53) under the joint randomization of the signs and the controller.

It remains to make the adversarial sequence deterministic. For a fixed sign vector $\boldsymbol{\xi} \in \{-1,1\}^T$, let

$$R_{\mathcal{A}}(\boldsymbol{\xi}) \triangleq \mathbb{E}_{\mathcal{A}}\left[\text{Reg}_T^{\text{SDAC}} \,\middle|\, \boldsymbol{\xi}\right].$$

The bound just proved states that the average of $R_{\mathcal{A}}(\boldsymbol{\xi})$ over the 2^T sign vectors is at least the right-hand side of (53). Hence at least one deterministic vector $\boldsymbol{\xi}^\star$ attains that value. Fixing $\boldsymbol{\xi}^\star$ in (60) produces a deterministic oblivious admissible sequence, and the only remaining expectation is over the internal randomization of $\mathcal{A}$. □

Proof of Corollary 1. The definition of $t_{\rm mix}(\alpha)$ gives

$$\alpha^{t_{\rm mix}(\alpha)} \le \alpha^{\log 2/\log(1/\alpha)} = \frac{1}{2}. \tag{76}$$

Therefore, for every $t \ge t_{\rm mix}(\alpha)+2$,

$$\begin{aligned}\beta_t &\ge \alpha\left(L_u + \frac{L_x}{2(1-\alpha)}\right)\\ &\ge \frac{\alpha}{2}\left(L_u + \frac{L_x}{1-\alpha}\right) = \frac{G_\alpha}{2}.\end{aligned} \tag{77}$$

There are $T - t_{\rm mix}(\alpha) - 1$ indices in $\{t_{\rm mix}(\alpha)+2,\dots,T\}$. If $T \ge 2t_{\rm mix}(\alpha)+2$, then

$$T - t_{\rm mix}(\alpha) - 1 \ge \frac{T}{2}. \tag{78}$$

Applying Theorem 2 and retaining only these indices,

$$\begin{aligned}\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\rm SDAC}\big] &\ge \frac{1}{2\sqrt{2}}\left[\frac{T}{2}\left(\frac{G_\alpha}{2}\right)^2\right]^{1/2}\\ &= \frac{G_\alpha}{8}\sqrt{T}.\end{aligned} \tag{79}$$

This is (56). □

We next prove the expressiveness result stated in Section IV-C.

Proof of Lemma 4. Fix $\mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}$ and set

$$a_i \triangleq \frac{q_i}{\alpha^i}, \qquad b_j \triangleq 1 - \sum_{i=1}^{j-1} a_i, \qquad j \ge 1.$$

The weighted-budget constraint gives $a_i \ge 0$ and $b_j \ge 0$. Moreover,

$$x_t(\mathbf{q}) = \sum_{j=1}^{t-1} \alpha^{j-1} b_j w_{t-j}. \tag{80}$$

Indeed, the identity is immediate at $t=1$. If it holds at time t, then

$$\begin{aligned}x_{t+1}(\mathbf{q}) &= w_t + \sum_{j=1}^{t-1}\big(\alpha^j b_j - q_j\big) w_{t-j}\\ &= w_t + \sum_{j=1}^{t-1} \alpha^j b_{j+1} w_{t-j},\end{aligned}$$

because $q_j = \alpha^j a_j$ and $b_{j+1} = b_j - a_j$. Reindexing gives (80) at time $t+1$.

For each $j \ge 1$,

$$a_j \le 1 - \sum_{i=1}^{j-1} a_i = b_j.$$

Consequently,

$$\begin{aligned}0 \le u_t(\mathbf{q}) &= \sum_{j=1}^{t-1} \alpha^j a_j w_{t-j}\\ &\le \sum_{j=1}^{t-1} \alpha^j b_j w_{t-j} = \alpha x_t(\mathbf{q}),\end{aligned}$$

which proves (2) for the decreasing-allocation policy.

For the truncated vector in (33),

$$m_{H,i}(\mathbf{q}) = a_i \ge 0, \qquad \sum_{i=1}^{H} m_{H,i}(\mathbf{q}) \le \sum_{i=1}^{\infty} a_i \le 1.$$

Thus $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. Its SDAC action is

$$u_t\big(\mathbf{m}_H(\mathbf{q})\big) = \sum_{i=1}^{\min\{H,t-1\}} q_i w_{t-i}.$$

It follows that

$$0 \le u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big) = \sum_{i=H+1}^{t-1} q_i w_{t-i} \le \mathrm{Tail}_H(\mathbf{q}),$$

where the middle sum is empty when $t \le H+1$. This proves (34).

Define

$$e_t^x \triangleq x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}), \qquad e_t^u \triangleq u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big).$$

The two state recursions give

$$e_{t+1}^x = \alpha e_t^x + e_t^u, \qquad e_1^x = 0.$$

Since $0 \le e_t^u \le \mathrm{Tail}_H(\mathbf{q})$,

$$0 \le e_t^x = \sum_{s=1}^{t-1} \alpha^{t-1-s} e_s^u \le \frac{\mathrm{Tail}_H(\mathbf{q})}{1-\alpha},$$

which proves (35). The Lipschitz property now yields

$$\begin{aligned}&\big|c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big)\\ &\quad - c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\big|\\ &\quad \le \left(L_u + \frac{L_x}{1-\alpha}\right)\mathrm{Tail}_H(\mathbf{q}).\end{aligned}$$

Finally,

$$\begin{aligned}\mathrm{Tail}_H(\mathbf{q}) &= \sum_{i=H+1}^{\infty} \alpha^i a_i\\ &\le \alpha^{H+1} \sum_{i=H+1}^{\infty} a_i \le \alpha^{H+1}.\end{aligned}$$

Summing the per-period cost bound proves (36).

It remains to identify proportional state-feedback policies within the decreasing-allocation policy class. For $k \in [0,\alpha]$, let $\lambda_k = \alpha - k$ and let $\mathbf{q}(k)$ be defined by (37). The sequence is nonnegative and decreasing. If $k=0$, then $\mathbf{q}(k) = \mathbf{0} \in \mathcal{Q}_\alpha^{\downarrow}$. If $k>0$, then $\lambda_k/\alpha < 1$ and

$$\sum_{i=1}^{\infty} \frac{q_i(k)}{\alpha^i} = \frac{k}{\alpha}\sum_{j=0}^{\infty}\left(\frac{\lambda_k}{\alpha}\right)^j = \frac{k}{\alpha - \lambda_k} = 1,$$

so again $\mathbf{q}(k) \in \mathcal{Q}_\alpha^{\downarrow}$. Substituting $u_t(k) = k x_t(k)$ into the dynamics and unrolling gives

$$u_t(k) = k\sum_{i=1}^{t-1} \lambda_k^{i-1} w_{t-i} = \sum_{i=1}^{t-1} q_i(k) w_{t-i} = u_t\big(\mathbf{q}(k)\big).$$

The two policies therefore have identical action and state trajectories. Their SDAC truncations also coincide coordinatewise, and the omitted geometric tail is

$$\mathrm{Tail}_H\big(\mathbf{q}(k)\big) = k \sum_{i=H+1}^{\infty} \lambda_k^{i-1} = \frac{k\lambda_k^H}{1-\lambda_k}.$$

Substitution in (36) proves (38). $\square$

The remaining lemmas establish the OMD and stability estimates used in the proof of Theorem 1; they rely on the state-decomposition, feasibility, and surrogate-cost lemmas already established in Section IV (Lemmas 1, 2, and 3).

Lemma 6 (Bregman projection onto $\mathcal{M}_H$)**.** *Let D_Ψ be the Bregman divergence* (40) *of the unnormalized negentropy* (39)*. Fix $\widetilde{\mathbf{m}} \in \mathbb{R}^H_{>0}$. Then the Bregman projection of $\widetilde{\mathbf{m}}$ onto $\mathcal{M}_H$,*

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) \triangleq \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}),$$

is given by

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\max\{1, \|\widetilde{\mathbf{m}}\|_1\}}.$$

Proof. The projection objective $\mathbf{m} \mapsto D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}})$ is strictly convex, and, for $m_i > 0$,

$$\frac{\partial}{\partial m_i} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}) = \log \frac{m_i}{\widetilde{m}_i}. \tag{81}$$

Its unconstrained minimizer is therefore $\mathbf{m} = \widetilde{\mathbf{m}}$.

If $\|\widetilde{\mathbf{m}}\|_1 \le 1$, this minimizer belongs to $\mathcal{M}_H$, and hence

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \widetilde{\mathbf{m}}.$$

Suppose $\|\widetilde{\mathbf{m}}\|_1 > 1$. The constraint $\|\mathbf{m}\|_1 \le 1$ is then active. Moreover, the minimizer has no zero coordinate: if $m_{i_0} = 0$, transferring an arbitrarily small amount of mass from a positive coordinate to coordinate i_0 has directional derivative $-\infty$, because

$$\lim_{z\downarrow 0} \log \frac{z}{\widetilde{m}_{i_0}} = -\infty.$$

Thus the nonnegativity constraints are inactive. With $\nu \ge 0$ denoting the multiplier of $\|\mathbf{m}\|_1 \le 1$, stationarity and complementary slackness give

$$\log \frac{m_i}{\widetilde{m}_i} + \nu = 0, \qquad \|\mathbf{m}\|_1 = 1. \tag{82}$$

The first relation in (82) implies

$$m_i = \widetilde{m}_i \exp(-\nu).$$

Summing over i and using the second relation in (82),

$$1 = \|\widetilde{\mathbf{m}}\|_1 \exp(-\nu), \qquad \exp(-\nu) = \frac{1}{\|\widetilde{\mathbf{m}}\|_1}.$$

Hence $\nu = \log \|\widetilde{\mathbf{m}}\|_1 > 0$ and

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\|\widetilde{\mathbf{m}}\|_1}.$$

Combining the cases $\|\widetilde{\mathbf{m}}\|_1 \le 1$ and $\|\widetilde{\mathbf{m}}\|_1 > 1$ proves the stated formula. $\square$

Lemma 7 (Closed-form solution of the OMD update)**.** *Fix $\mathbf{m}_t \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$, $\eta > 0$, and $\mathbf{g}_t \in \mathbb{R}^H$, and let $\widetilde{\mathbf{m}}_{t+1}$ be defined coordinatewise by* (49)*, i.e. $\widetilde{m}_{t+1,i} = m_{t,i}\exp(-\eta g_{t,i})$. Then*

$$\begin{aligned}
&\operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} \Big\{ \eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t) \Big\} \\
&\quad = \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) = \mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}_{t+1}),
\end{aligned}$$

where $\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}$ is the Bregman projection of Lemma 6. Consequently, the OMD update (48) *coincides exactly with the closed-form OMD update* (49)–(50)*, and in particular $\mathbf{m}_{t+1} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$. Thus the positive initialization* (41) *keeps every algorithm iterate strictly positive.*

Proof. Fix $\mathbf{m} \in \mathbb{R}^H_{\ge 0}$ and a coordinate i. Then

$$\begin{aligned}
\eta g_{t,i}\, m_i + m_i \log \frac{m_i}{m_{t,i}} &= m_i \left(\eta g_{t,i} + \log \frac{m_i}{m_{t,i}} \right) \\
&= m_i \log \frac{m_i}{\widetilde{m}_{t+1,i}}.
\end{aligned} \tag{83}$$

Substitution into the OMD objective gives

$$\begin{aligned}
&\eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t) \\
&\quad = D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) + \sum_{i=1}^{H} \left(m_{t,i} - \widetilde{m}_{t+1,i} \right).
\end{aligned} \tag{84}$$

The final sum is independent of $\mathbf{m}$. Hence the two optimization problems have the same minimizer, and Lemma 6 gives (50). Since $\mathbf{m}_t > 0$ coordinatewise, both the multiplicative step and the subsequent positive rescaling preserve strict positivity. $\square$

Lemma 8 (One-step movement of OMD)**.** *For the OMD update with step size $\eta > 0$ and subgradient $\mathbf{g}_t \in \partial \ell_t(\mathbf{m}_t)$,*

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \le \eta \, \|\mathbf{g}_t\|_\infty.$$

In particular, since $\|\mathbf{g}_t\|_\infty \le G_\alpha$, $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \le \eta G_\alpha$.

Proof. First-order optimality of (48), evaluated at the feasible point $\mathbf{m}_t$, gives

$$\begin{aligned}
&\langle \eta\mathbf{g}_t + \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \ge 0, \\
&D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) + D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1}) \\
&\qquad \le \eta\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \\
&\qquad \le \eta\|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.
\end{aligned} \tag{85}$$

The second line uses the symmetric Bregman identity.

For every $\mathbf{z} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$ and $\mathbf{v} \in \mathbb{R}^H$,

$$\begin{aligned}
\mathbf{v}^\top \nabla^2 \Psi(\mathbf{z}) \mathbf{v} &= \sum_{i=1}^{H} \frac{v_i^2}{z_i} \\
&\ge \frac{\|\mathbf{v}\|_1^2}{\sum_{i=1}^{H} z_i} \\
&\ge \|\mathbf{v}\|_1^2.
\end{aligned} \tag{86}$$

The first inequality is Cauchy–Schwarz. The segment joining $\mathbf{m}_t$ and $\mathbf{m}_{t+1}$ lies in $\mathcal{M}_H \cap \mathbb{R}^H_{>0}$ by Lemma 7; integrating (86) along this segment yields

$$D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) \geq \frac{1}{2}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2,$$
$$D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1}) \geq \frac{1}{2}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2.$$

Combining these inequalities with (85),

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2 \leq \eta\|\mathbf{g}_t\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.$$

The claim is immediate if $\mathbf{m}_{t+1} = \mathbf{m}_t$; otherwise divide by $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1$. □

Lemma 9 (OMD regret for the surrogate losses)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the update* (48) *satisfies*

$$\sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) \leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2. \quad (87)$$

Proof. Convexity of ℓ_t gives

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\leq \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}\rangle \\ &= \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle.\end{aligned} \quad (88)$$

First-order optimality of (48), evaluated at $\mathbf{m} \in \mathcal{M}_H$, yields

$$\begin{aligned}\eta\langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle &\leq \langle \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m} - \mathbf{m}_{t+1}\rangle \\ &= D_\Psi(\mathbf{m}\|\mathbf{m}_t) - D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1}) \\ &\quad - D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t).\end{aligned} \quad (89)$$

The equality is the three-point Bregman identity. Substituting (89) into (88),

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_t)}{\eta} - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})}{\eta} \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \\ &\quad - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta}.\end{aligned} \quad (90)$$

The strong-convexity calculation (86) and Hölder's inequality give

$$\begin{aligned}&\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta} \\ &\quad \leq \|\mathbf{g}_t\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 - \frac{\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2}{2\eta} \\ &\quad \leq \frac{\eta}{2}\|\mathbf{g}_t\|_\infty^2.\end{aligned} \quad (91)$$

The last inequality is Young's inequality. Combining (90) and (91), and then summing over t, gives

$$\begin{aligned}\sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) &\leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} \\ &\quad - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1})}{\eta} \\ &\quad + \frac{\eta}{2}\sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2.\end{aligned}$$

Since $D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1}) \geq 0$, this proves (87). □

Lemma 10 (Positive-part representation of the dynamics)**.** *For $\mathbf{m} \in \mathcal{M}_H$, define*

$$F_t(x; \mathbf{m}) \triangleq [\alpha x - \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle]_+ + w_t. \quad (92)$$

Then the online state and every fixed-policy state satisfy

$$x_{t+1} = F_t(x_t; \mathbf{m}_t), \qquad x_{t+1}(\mathbf{m}) = F_t(x_t(\mathbf{m}); \mathbf{m}). \quad (93)$$

Moreover, for all $x, x' \geq 0$ and $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$\begin{aligned}&|F_t(x; \mathbf{m}) - F_t(x'; \mathbf{m}')| \\ &\quad \leq |\alpha(x - x') - \langle \boldsymbol{\phi}_t^u, \mathbf{m} - \mathbf{m}'\rangle| \\ &\quad \leq \alpha|x - x'| + \|\boldsymbol{\phi}_t^u\|_\infty\|\mathbf{m} - \mathbf{m}'\|_1.\end{aligned} \quad (94)$$

Proof. For the online trajectory, (43) implies

$$\begin{aligned}\alpha x_t - u_t &= \alpha x_t - \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\} \\ &= [\alpha x_t - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle]_+ .\end{aligned} \quad (95)$$

For a fixed $\mathbf{m}$, Lemma 2 gives

$$\alpha x_t(\mathbf{m}) - \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle \geq 0,$$

so inserting a positive part does not change its state update. This proves (93). Finally, the map $z \mapsto [z]_+$ is nonexpansive:

$$|[z]_+ - [z']_+| \leq |z - z'|.$$

Apply this inequality to the two arguments in (92), and then apply Hölder's inequality, to obtain (94). □

Lemma 11 (Contractive state and action deviation)**.** *For the online trajectory and the counterfactual fixed-policy trajectory associated with the current internal iterate, one has, for every $t \in [T]$,*

$$|x_t - x_t(\mathbf{m}_t)| \leq \frac{\alpha\eta G_\alpha}{(1-\alpha)^2}, \quad (96)$$

$$0 \leq u_t(\mathbf{m}_t) - u_t \leq \frac{\alpha^2\eta G_\alpha}{(1-\alpha)^2}. \quad (97)$$

Here $x_t(\mathbf{m}_t)$ is generated by using the single fixed parameter $\mathbf{m}_t$ from time 1 through time t.

Proof. Define

$$d_t \triangleq |x_t - x_t(\mathbf{m}_t)|.$$

Since $x_1 = x_1(\mathbf{m}_1) = 0$, one has $d_1 = 0$. For $t \in \{1, \ldots, T-1\}$, Lemma 10 gives

$$\begin{aligned}d_{t+1} &= |F_t(x_t; \mathbf{m}_t) - F_t(x_t(\mathbf{m}_{t+1}); \mathbf{m}_{t+1})| \\ &\leq \alpha\,|x_t - x_t(\mathbf{m}_{t+1})| + \|\boldsymbol{\phi}_t^u\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.\end{aligned} \quad (98)$$

The affine state representation (23) implies

$$\begin{aligned}|x_t(\mathbf{m}_t) - x_t(\mathbf{m}_{t+1})| &= |\langle \boldsymbol{\phi}_t^x, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle| \\ &\leq \|\boldsymbol{\phi}_t^x\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.\end{aligned} \quad (99)$$

Combining the triangle inequality with (98) and (99) yields

$$\begin{aligned}d_{t+1} &\leq \alpha d_t \\ &\quad + (\alpha\|\boldsymbol{\phi}_t^x\|_\infty + \|\boldsymbol{\phi}_t^u\|_\infty)\,\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.\end{aligned} \quad (100)$$

By (26),

$$\begin{aligned}\alpha\|\boldsymbol{\phi}_t^x\|_\infty + \|\boldsymbol{\phi}_t^u\|_\infty &\leq \frac{\alpha^2}{1-\alpha} + \alpha \\ &= \frac{\alpha}{1-\alpha}.\end{aligned} \quad (101)$$

Lemma 8 and $\|\mathbf{g}_t\|_\infty \le G_\alpha$ therefore reduce (100) to

$$d_{t+1} \le \alpha d_t + \frac{\alpha\eta G_\alpha}{1-\alpha}. \tag{102}$$

Unrolling from $d_1 = 0$ gives

$$\begin{aligned} d_t &\le \frac{\alpha\eta G_\alpha}{1-\alpha}\sum_{j=0}^{t-2}\alpha^j \\ &\le \frac{\alpha\eta G_\alpha}{(1-\alpha)^2}, \end{aligned} \tag{103}$$

with an empty sum when $t = 1$. This proves (96).

Finally, $\widehat{u}_t = u_t(\mathbf{m}_t)$ and the feasibility of fixed SDAC policies imply

$$\begin{aligned} 0 \le u_t(\mathbf{m}_t) - u_t &= [u_t(\mathbf{m}_t) - \alpha x_t]_+ \\ &\le [\alpha x_t(\mathbf{m}_t) - \alpha x_t]_+ \\ &\le \alpha d_t. \end{aligned} \tag{104}$$

Combining (104) with (96) proves (97). □

Lemma 12 (Stability term)**.** *For all horizons T,*

$$\sum_{t=1}^{T}\Big(c_t(x_t,u_t) - \ell_t(\mathbf{m}_t)\Big) \le \frac{\alpha\eta T G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}.$$

Proof. Both state–action pairs lie in $\mathcal{D}_\alpha$. Hence (5) and Lemma 11 give

$$\begin{aligned} &c_t(x_t,u_t) - \ell_t(\mathbf{m}_t) \\ &\quad\le L_x\big|x_t - x_t(\mathbf{m}_t)\big| + L_u\big|u_t - u_t(\mathbf{m}_t)\big| \\ &\quad\le \frac{\alpha\eta G_\alpha L_x}{(1-\alpha)^2} + \frac{\alpha^2\eta G_\alpha L_u}{(1-\alpha)^2} \\ &\quad= \frac{\alpha\eta G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}. \end{aligned} \tag{105}$$

Summing (105) over $t = 1, \ldots, T$ proves the claim. □

Proof of Theorem 1. Define the stability term

$$\mathrm{Stab}_T \triangleq \sum_{t=1}^{T}\left(c_t(x_t,u_t) - \ell_t(\mathbf{m}_t)\right).$$

Using $\ell_t(\mathbf{m}) = c_t(x_t(\mathbf{m}), u_t(\mathbf{m}))$, the regret decomposes as

$$\mathrm{Reg}_T^{\mathrm{SDAC}} = \mathrm{Stab}_T + \sum_{t=1}^{T}\ell_t(\mathbf{m}_t) - \min_{\mathbf{m}\in\mathcal{M}_H}\sum_{t=1}^{T}\ell_t(\mathbf{m}). \tag{106}$$

We first bound the initial Bregman divergence. Fix $\mathbf{m} \in \mathcal{M}_H$. Since $\mathbf{m}_1 = \frac{1}{H+1}\mathbf{1}$,

$$\begin{aligned} D_\Psi(\mathbf{m}\|\mathbf{m}_1) &= \sum_{i=1}^{H} m_i \log\big((H+1)m_i\big) \\ &\quad - \|\mathbf{m}\|_1 + \frac{H}{H+1} \\ &= \sum_{i=1}^{H} m_i\log m_i + \|\mathbf{m}\|_1\log(H+1) \\ &\quad - \|\mathbf{m}\|_1 + \frac{H}{H+1}. \end{aligned} \tag{107}$$

Because $z\log z \le 0$ for $z \in [0,1]$,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \le \|\mathbf{m}\|_1\log(H+1) - \|\mathbf{m}\|_1 + \frac{H}{H+1}. \tag{108}$$

The right-hand side is affine in $\|\mathbf{m}\|_1 \in [0,1]$, and its values at the endpoints are

$$\begin{aligned} \frac{H}{H+1} &\le \log(H+1), \\ \log(H+1) - \frac{1}{H+1} &\le \log(H+1). \end{aligned}$$

Here the first inequality follows from $\log z \ge 1 - z^{-1}$ for $z \ge 1$. Therefore,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \le \log(H+1) \qquad \text{for every } \mathbf{m} \in \mathcal{M}_H. \tag{109}$$

Lemma 9, $\|\mathbf{g}_t\|_\infty \le G_\alpha$, and (109) imply, for every $\mathbf{m} \in \mathcal{M}_H$,

$$\begin{aligned} \sum_{t=1}^{T}\big(\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\big) &\le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2 \\ &\le \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}. \end{aligned} \tag{110}$$

Since (110) holds uniformly over $\mathbf{m}$,

$$\sum_{t=1}^{T}\ell_t(\mathbf{m}_t) - \min_{\mathbf{m}\in\mathcal{M}_H}\sum_{t=1}^{T}\ell_t(\mathbf{m}) \le \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}. \tag{111}$$

Lemma 12 gives

$$\mathrm{Stab}_T \le \frac{\alpha\eta T G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}. \tag{112}$$

Substituting (111) and (112) into (106),

$$\begin{aligned} \mathrm{Reg}_T^{\mathrm{SDAC}} &\le \frac{\log(H+1)}{\eta} + \frac{\eta T G_\alpha^2}{2} \\ &\quad + \frac{\alpha\eta T G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2} \\ &= \frac{\log(H+1)}{\eta} + \eta T\Gamma_\alpha. \end{aligned} \tag{113}$$

Differentiating with respect to η yields

$$-\frac{\log(H+1)}{\eta^2} + T\Gamma_\alpha.$$

The unique minimizer is therefore

$$\eta^* = \sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}. \tag{114}$$

At this value the two summands in (113) are equal, so

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le 2\sqrt{\Gamma_\alpha T\log(H+1)}. \tag{115}$$

It remains to verify the dependence on α. As $\alpha \downarrow 0$,

$$\begin{aligned} \frac{G_\alpha}{\alpha} &= \frac{L_x}{1-\alpha} + L_u \longrightarrow L_x + L_u, \\ \frac{\Gamma_\alpha}{\alpha^2} &= \frac{1}{2}\left(\frac{G_\alpha}{\alpha}\right)^2 + \frac{G_\alpha}{\alpha}\frac{L_x+\alpha L_u}{(1-\alpha)^2} \\ &\longrightarrow \frac{(L_x+L_u)^2}{2} + L_x(L_x+L_u) > 0. \end{aligned}$$

Therefore $\sqrt{\Gamma_\alpha} = \Theta(\alpha)$ as $\alpha \downarrow 0$. As $\alpha \uparrow 1$, write $\varepsilon = 1 - \alpha$. Then

$$\begin{aligned}\varepsilon G_\alpha &= \alpha(L_x + \varepsilon L_u) \longrightarrow L_x,\\ \varepsilon^3 \Gamma_\alpha &= \frac{\varepsilon^3 G_\alpha^2}{2} + \alpha(\varepsilon G_\alpha)(L_x + \alpha L_u)\\ &\longrightarrow L_x(L_x + L_u) > 0.\end{aligned}$$

Thus $\sqrt{\Gamma_\alpha} = \Theta((1-\alpha)^{-3/2})$ as $\alpha \uparrow 1$. □

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